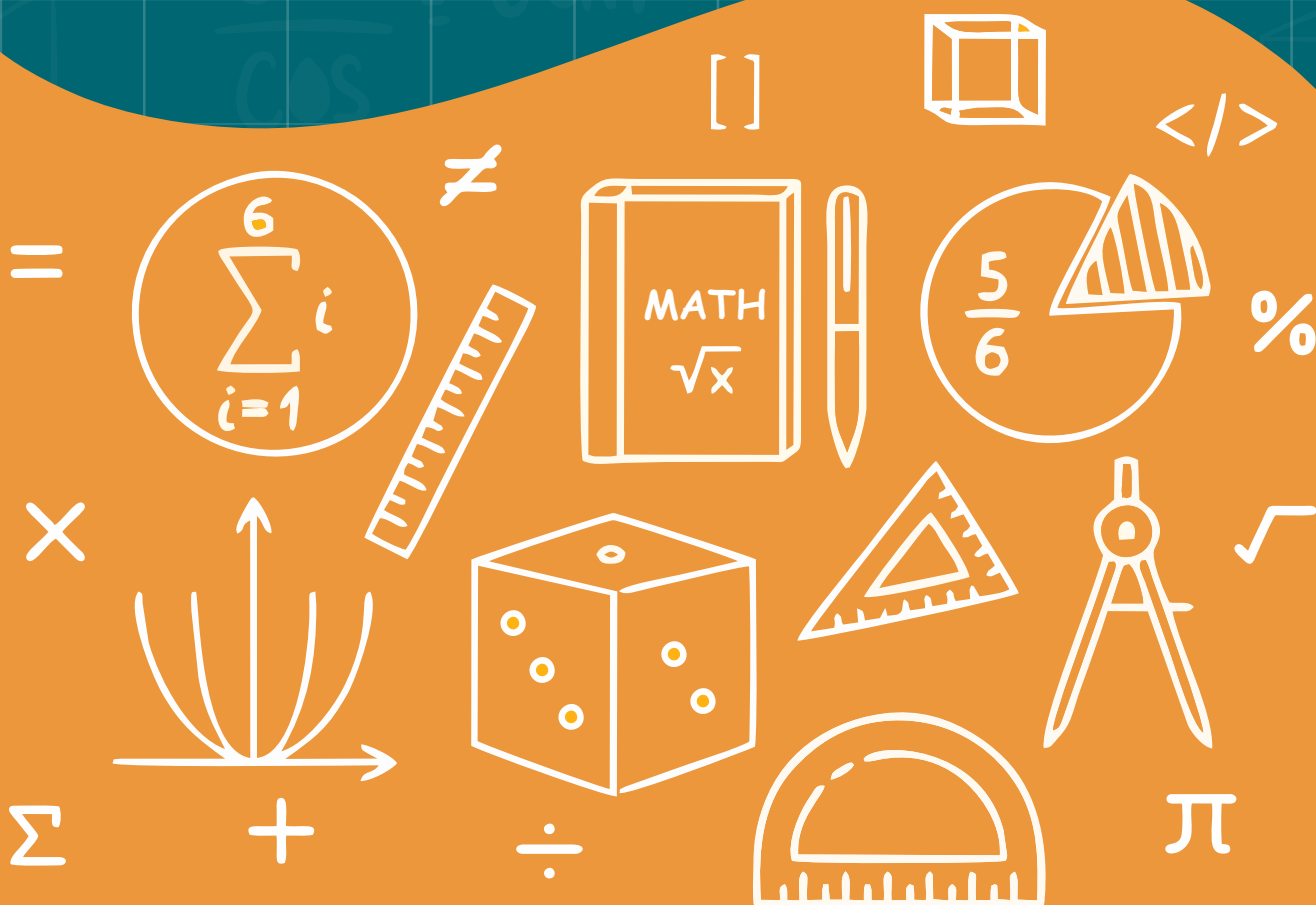


UNIT 5

SIMPLIFICATION



The process of getting a simplified number is known as simplification.

KINDS OF BRACKETS

There are four kinds of brackets.

- (i) _____ Vinculum or bar
- (ii) () parenthesis or curved brackets or round brackets
- (iii) { } braces or curly brackets
- (iv) [] square brackets

Know the order of preference as, $-$, $()$, $\{ \}$ and $[]$, to remove (simplify) them from an expression.

We use following rules to remove the brackets.

(i) If there is a plus (+) sign before brackets, the brackets are removed without changing the sign of the number within brackets.

For example, $(2 - 5) = (-3) = -3$

(ii) If there is a minus' sign before brackets, the brackets are removed and the sign of the number within brackets is changed.

For example, $-(2 - 5) = -(-3) = 3$

(iii) If there is a number before brackets, the number in the brackets left after simplification is multiplied by this number and the brackets are removed.

For example, $4(2 - 5) = 4(-3) = -12$

Also $-5(3 - 7) = -5(-4) = 20$

BODMAS RULE

BODMAS indicates the sequence in which more than one operations are performed in the simplification process of any mathematical problem.

The chain of letters in BODMAS shows meaning of each letter as:

- B** for "Brackets"
- O** for "order of operation"
- D** for "Division"
- M** for "Multiplication"
- A** for "Addition"
- S** for "Subtraction"

Removal of brackets and solving of operation 'of' with operations of DMAS is known as BODMAS rule.

Example 1: Simplify: $5 + \{16 - 4 + 2 \times 3 - (-6 \div 2)\}$

Solution:

$$\begin{aligned}
 & 5 + \{16 - 4 + 2 \times 3 - (-6 \div 2)\} \\
 &= 5 + \{16 - 4 + 6 - (-3)\} \\
 &= 5 + \{16 + 2 + 3\} \\
 &= 5 + (18 + 3) \\
 &= 5 + 21 = 26
 \end{aligned}$$

Example 2: Simplify: $2 [-2 \{4 \times 9 \div 8 (5 - 3 - 4)\}]$

Solution:

$$\begin{aligned}
 & 2 [-2 \{4 \times 9 \div 8 (5 - 3 - 4)\}] \\
 &= 2 [-2 \{4 \times 9 \div 8 (5 - (-1))\}] \\
 &= 2 [-2 \{4 \times 9 \div 8 (5 + 1)\}] \\
 &= 2 [-2 \{4 \times 9 \div (8 \times 6)\}]
 \end{aligned}$$

$$= 2 [-2 \{4 \times 9 \div 48\}]$$

$$= 2 [-2 \{ \overset{1}{\cancel{4}} \times \overset{3}{\cancel{9}} \}]$$

$\cancel{48}^{12^4}$

$$= 2 [-2 \{1 \times 3\}]$$

4

$$= \overset{1}{\cancel{2}} [-\overset{1}{\cancel{2}} \times 3]$$

$\cancel{4}^{2^1}$

$$= -1 \times 3 = -3$$

Simplify mathematical expressions involving fractions and decimals grouped with brackets using BODMAS rule:

Let us solve some examples using BODMAS rule.

Example 1: Simplify:

$$3 \frac{1}{4} \div \left[\frac{2}{5} + \left\{ \left(\frac{7}{8} - \frac{7}{12} \right) + 1 \frac{5}{6} \right\} - 2 \frac{2}{5} \right]$$

Solution:

$$\begin{aligned} & 3 \frac{1}{4} \div \left[\frac{2}{5} + \left\{ \left(\frac{7}{8} - \frac{7}{12} \right) + 1 \frac{5}{6} \right\} - 2 \frac{2}{5} \right] \\ &= \frac{13}{4} \div \left[\frac{2}{5} + \left\{ \left(\frac{7}{8} - \frac{7}{12} \right) + \frac{11}{6} \right\} - \frac{12}{5} \right] \quad \text{(Converting mixed fractions into improper fractions)} \\ &= \frac{13}{4} \div \left[\frac{2}{5} + \left\{ \left(\frac{21 - 14}{24} \right) + \frac{11}{6} \right\} - \frac{12}{5} \right] \quad \text{(Simplifying within parenthesis)} \\ &= \frac{13}{4} \div \left[\frac{2}{5} + \left\{ \frac{7}{24} + \frac{11}{6} \right\} - \frac{12}{5} \right] \quad \text{(Removing parenthesis)} \end{aligned}$$

$$\begin{aligned}
&= \frac{13}{4} \div \left[\frac{2}{5} + \left\{ \frac{7+44}{24} \right\} - \frac{12}{5} \right] \text{ (Simplifying within braces)} \\
&= \frac{13}{4} \div \left[\frac{2}{5} + \frac{51}{24} - \frac{12}{5} \right] \text{ (Removing braces)} \\
&= \frac{13}{4} \div \left[\frac{48+255-288}{120} \right] \text{ (Simplifying within square brackets)} \\
&= \frac{13}{4} \div \frac{15}{120} \text{ (Removing square brackets)} \\
&= \frac{13}{\cancel{4}^1} \times \frac{\cancel{120}^{30^2}}{\cancel{15}^1} = 13 \times 2 \\
&= 26
\end{aligned}$$

Example 2: Simplify:

$$1 \frac{3}{4} \div \left[9 \frac{5}{8} \div \left\{ \frac{2}{5} + \left(8 \frac{2}{5} \times \overline{4 \frac{1}{6} - 3 \frac{1}{2}} \right) \right\} \right]$$

Solution:

$$\begin{aligned}
&1 \frac{3}{4} \div \left[9 \frac{5}{8} \div \left\{ \frac{2}{5} + \left(8 \frac{2}{5} \times \overline{4 \frac{1}{6} - 3 \frac{1}{2}} \right) \right\} \right] \\
&= \frac{7}{4} \div \left[\frac{77}{8} \div \left\{ \frac{2}{5} + \left(\frac{42}{5} \times \overline{\frac{25}{6} - \frac{7}{2}} \right) \right\} \right] \text{ (Converting mixed fractions into improper fractions)} \\
&= \frac{7}{4} \div \left[\frac{77}{8} \div \left\{ \frac{2}{5} + \left(\frac{42}{5} \times \overline{\frac{25-21}{6}} \right) \right\} \right] \text{ (Solving vinculum)} \\
&= \frac{7}{4} \div \left[\frac{77}{8} \div \left\{ \frac{2}{5} + \left(\frac{\cancel{42}^7}{5} \times \frac{\cancel{4}}{\cancel{6}^1} \right) \right\} \right] \text{ (Simplifying within parenthesis)}
\end{aligned}$$

$$= \frac{7}{4} \div \left[\frac{77}{8} \div \left\{ \frac{2}{5} + \frac{28}{5} \right\} \right] \text{ (Removing parenthesis)}$$

$$= \frac{7}{4} \div \left[\frac{77}{8} \div \left\{ \frac{\cancel{30}^6}{\cancel{5}_1} \right\} \right] \text{ (Solving braces)}$$

$$= \frac{7}{4} \div \left[\frac{77}{8} \times \frac{1}{6} \right] \text{ (Removing braces)}$$

$$= \frac{7}{4} \div \frac{77}{48} = \frac{\cancel{7}^1}{\cancel{4}_1} \times \frac{\cancel{48}^{12}}{\cancel{77}^{11}} = \frac{12}{11} = 1 \frac{1}{11}$$

Example 3: Simplify:

$$6.4 - \overline{(3.3 - 1.2 \times 2.5)}$$

Solution:

$$6.4 - \overline{(3.3 - 1.2 \times 2.5)} \text{ (Solving vinculum)}$$

$$= 6.4 - (2.1 \times 2.5) \text{ (Simplifying within parenthesis)}$$

$$= 6.4 - 5.25 \text{ (Removing parenthesis)}$$

$$= 1.15$$

Example 4: Simplify:

$$2.9 \times \overline{1.8 - 1.3} \times 3.75$$

Solution:

$$2.9 \times \overline{1.8 - 1.3} \times 3.75$$

$$= 2.9 \times 0.5 \times 3.75$$

$$= 5.4375 \text{ or } 5.44$$

EXERCISE 5.1

Q1. Solve the following:

i	$3(7 + 2)$	
ii	$5(6 - 4)$	
iii	$2(10 - 2 \times 3)$	
iv	$21 - 3(4 - 1)$	
v	$2(2 + 3) - 7$	
vi	$3 \times 4 - 2(5 - 2)$	
vii	$7 \times 3 - 5(4 - 2)$	
viii	$5 \times 4 - 2 \times 6$	
ix	$4 \times 9 - 2 \times 8$	
x	$4(5 \times 3 - 10)$	

Q2. Solve the following by using BODMAS rule:

(i) $3 \times 3 + (26 \times 4) + 19 + 27 =$

(ii) $2 + 23 + 28 \times 3 - (18 \div 29) =$

(iii) $20 + 40 + (20 \times 10) \div 5 \times 5 =$

(iv) $29 - 15 + 3 + (31 \times 2 \div 28) =$

(v) $1 \times 2 \times 6 \times (7 + 19 + 21) =$

Q3. Simplify the following:

(i) $\frac{1}{5} - \left\{ \frac{2}{5} \div \frac{4}{15} \times \frac{1}{2} - \left(\frac{1}{2} + \frac{2}{5} \right) \right\}$

$$(ii) \quad \frac{3}{4} + \left(\frac{4}{3} - \overline{\frac{1}{2} + \frac{2}{3} + \frac{4}{5}} \right)$$

$$(iii) \quad \frac{7}{8} + \left\{ \frac{5}{7} + \left(\frac{1}{2} + \frac{1}{3} \right) \div \frac{5}{6} \right\}$$

$$(iv) \quad \left(\overline{\frac{2}{3} - \frac{3}{4}} \times \frac{1}{8} \right) \div \left(\frac{2}{3} - \overline{\frac{3}{4} + \frac{5}{8}} \right)$$

$$(v) \quad 1\frac{1}{2} + \left[\frac{1}{3} \div \left\{ \frac{2}{5} \div \left(\frac{4}{5} - \overline{\frac{1}{10} + \frac{3}{5}} \right) \right\} \right]$$

$$(vi) \quad \left[\frac{1}{2} + \left\{ \left(\frac{3}{4} \div \frac{9}{16} \right) \times 1\frac{1}{2} \right\} - 2\frac{1}{4} \right]$$

$$(vii) \quad \left[\left(2\frac{3}{4} + 1\frac{2}{6} \right) \times \left(1\frac{1}{5} \div \frac{2}{5} \right) \right]$$

$$(viii) \quad \left[2\frac{1}{4} \times \left\{ 3\frac{1}{3} - \frac{1}{2} + \left(\frac{5}{3} - \frac{1}{6} \right) \right\} \right] + \frac{5}{6}$$

$$(ix) \quad \left(3\frac{1}{2} \times 5\frac{1}{3} + 10\frac{2}{5} \right) \times \left(3\frac{1}{8} - 6\frac{1}{2} \div 2\frac{1}{6} \right)$$

$$(x) \quad 4.02 \times \{1.2 \times \overline{2.11 + 3.89}\}$$

$$(xi) \quad 5.7 \times \{6.7 + \overline{(1.3 \times 7.2 \div 1.5 \times 2.4)}\}$$

WORD PROBLEMS

Example 1.

The maximum marks of Urdu test are 40 and passing marks are quarter of maximum marks. How many marks at least should Haris obtain to pass the test?

Solution:

$$\begin{aligned}\text{Maximum marks} &= 40 \\ \text{Maximum passing marks} &= 1 \text{ of } \frac{40}{4} \\ &= 1 \times 40 \frac{1}{4} \\ &= 10\end{aligned}$$

Example 2.

Fouzia gave $\frac{1}{4}$ of her cake to her sister, another $\frac{2}{3}$ to her two brothers and kept the rest of the cake for herself. What fraction of the cake was left with her.

Solution:

She gave $\frac{1}{4}$ and $\frac{2}{3}$ of the cake to her sister and two brothers respectively.

$$\text{Total cake given} = \frac{1}{4} + \frac{2}{3}$$

$$\text{Cake left with Fouzia} = 1 - \left(\frac{1}{4} + \frac{2}{3} \right) = 1 - \left(\frac{3}{12} + \frac{8}{12} \right)$$

$$= 1 - \frac{11}{12} = \frac{12}{12} - \frac{11}{12} = \frac{1}{12}$$

Hence $\frac{1}{12}$ th part of the cake was left with Fouzia.

EXERCISE 5.2

1. There are 640 students in total in a school. Two of the $\frac{3}{10}$ students use their own vehicles, $\frac{5}{8}$ use public transport and the remaining students come on foot. How many students of the school come on foot?
2. From a box of 140 oranges, $\frac{3}{7}$ of the oranges were sold 14 got spoilt. What is the number of the oranges left in the box?
3. A crowd of people was gathered for a cricket match. $\frac{1}{8}$ of crowd had come by train, $\frac{2}{7}$ by bus, $\frac{1}{4}$ by own vehicles and the remaining people came on foot. Find the part of people who came on foot.
4. Maira had Rs 58,000. She gave $\frac{16}{29}$ of it to her mother and $\frac{1}{4}$ of the remainder in Zakat fund. Find the amount left with her?
5. Aamir planted 425 plants of tomatoes. After some days 25 plants died. What fraction of plants was survived?
6. Aijaz gave $\frac{2}{5}$ of his money to his daughter, to his son $\frac{1}{7}$ and the remainder to his wife. What fraction of the amount was given to his wife.